

Question 1

How would you prevent two appointments for the same doctor from overlapping? Outline the constraint or trigger logic.

Solution:

To prevent overlapping time slots, a **BEFORE INSERT** and **BEFORE UPDATE** trigger is the most reliable method in MySQL. The logic checks if a new appointment's time range conflicts with any existing record for that specific doctor.

SQL

DELIMITER \$\$

CREATE TRIGGER check_appointment_overlap

BEFORE INSERT ON Appointment

FOR EACH ROW

BEGIN

 DECLARE overlap_count INT;

 -- Check if the doctor already has an appointment during this time range

 SELECT COUNT(*) INTO overlap_count

 FROM Appointment

 WHERE Doctor_ID = NEW.Doctor_ID

 AND NEW.Date = Date -- Same day

 AND (

 (NEW.Start_Time >= Start_Time AND NEW.Start_Time < End_Time) OR -- New starts during old

 (NEW.End_Time > Start_Time AND NEW.End_Time <= End_Time) OR -- New ends during old

 (NEW.Start_Time <= Start_Time AND NEW.End_Time >= End_Time) -- New spans over old

);

```
-- If an overlap exists, block the insert and throw an error

IF overlap_count > 0 THEN

    SIGNAL SQLSTATE '45000'

    SET MESSAGE_TEXT = 'Error: This doctor already has an overlapping appointment
scheduled at this time.';

END IF;

END$$

DELIMITER ;
```

Question 2

Describe a query to fetch each patient's plus their .

Solution (Fetching Patients with their Total Appointment Count):

Assuming it wants to know a patient's total history count within the system, we aggregate the appointments using a LEFT JOIN so even patients with zero appointments show up cleanly:

SQL

SELECT

Patient.Patient_ID,

Patient.Name AS Patient_Name,

COUNT(Appointment.Appointment_ID) AS Total_Appointments

FROM Patient

LEFT JOIN Appointment ON Patient.Patient_ID = Appointment.Patient_ID

GROUP BY Patient.Patient_ID, Patient.Name;

Question 3

You need a view showing each doctor's upcoming appointment count for next month. What would its GROUP BY look like?

Solution:

To count appointments per doctor, your GROUP BY clause must aggregate records using the doctor's primary identifier and their name.

The structural SQL code for that specific view looks like this:

SQL

```
CREATE VIEW View_NextMonthDoctorAppointments AS
```

```
SELECT
```

```
    Doctor.Doctor_ID,
```

```
    Doctor.Name AS Doctor_Name,
```

```
    COUNT(Appointment.Appointment_ID) AS Upcoming_Count
```

```
FROM Doctor
```

```
INNER JOIN Appointment ON Doctor.Doctor_ID = Appointment.Doctor_ID
```

```
WHERE Appointment.Date BETWEEN DATE_ADD(LAST_DAY(CURDATE()), INTERVAL 1 DAY)
```

```
        AND LAST_DAY(DATE_ADD(CURDATE(), INTERVAL 1 MONTH))
```

```
GROUP BY Doctor.Doctor_ID, Doctor.Name;
```